Virtrade

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Project Pitch

Paper Trading Website which allows you to see effectiveness of trading techniques and practice them to compete with other users.

Description of Project

- Practice stock trading without the possibility of losing money
- Uses real, current-day stock data and news stories to simulate actual conditions
- Can compare your results to other users in a leaderboard.

Project Description Continued

Project Tracker and VCS Repository

- Used GitHub for Projects Board Tracking and Project Management

- Two Code Reviews Prior for PR's
  Daily Management of Board

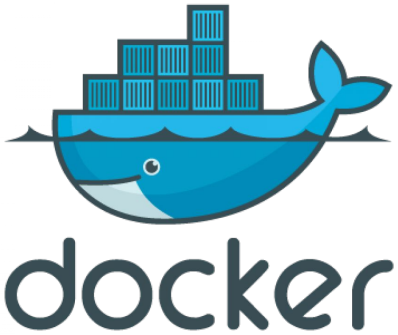
- 5/5 Stars

IDE

- VSCode
- GitHub PR integration
- Debugger
- 5/5 stars

Containerized Test Environment

- Docker volume with two services
- API Service, DB Service
- Standardized test environment across dev team’s machines
- 5/5 stars!



Database Management

- Database written via Postgress SQL
- New tables don’t create new dependencies on old tables
- Used user objects for lower maintenance needs such as stock buy and sell features
- 4/5 Stars

UI Tools

- HTML/Handlebars for UI
- Reusable and modular(Handle Bars)
- Frequent connections to database to display data, done with handlebars
- 4/5 stars

Test Suite

- Mocha test suite
- Unit tests for account registration validity
- ⅗ stars

Application Server

- Node JS handles running of application, including page rendering & database manipulation
- Flask API server for communication with external APIs and Postgres

- 4/5 stars

- Mostly composed of GET and POST methods, with auth

Deployable Environment

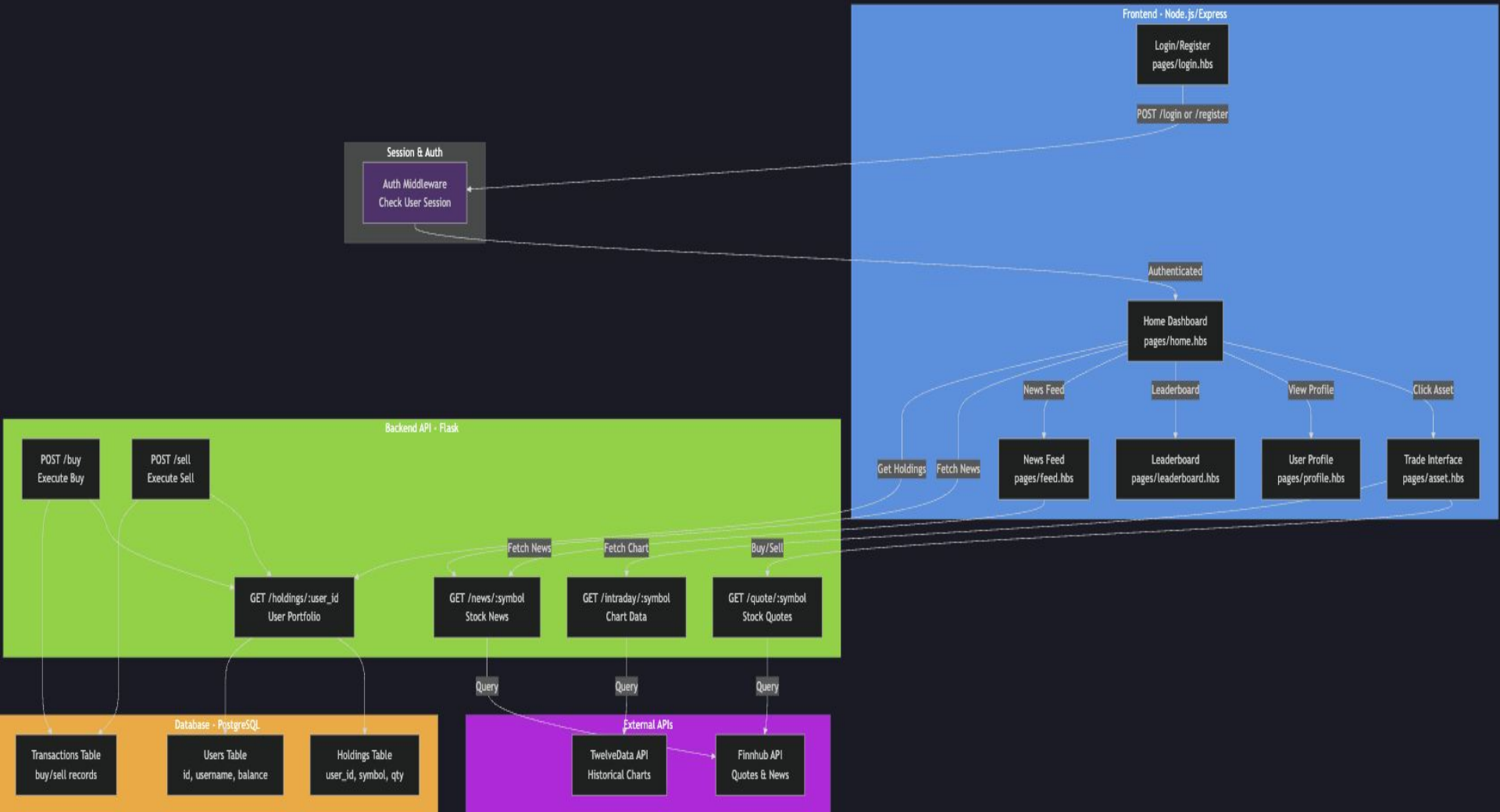
- Render has all API keys and cleanly hosts website
- 3/5 stars
- All API keys needs are done via Render to maintain scalability

Render

API

- Finnhub and Twelve Data for stock & news information.
- 4/5 stars
- In python file, parse API information and render it on given feed/home page

Finnhub


Frontend - Node.js/Express
Login/Register
pages/login.hbs
POST /login or /register
Session & Auth
Auth Middleware
Check User Session
Authenticated
Home Dashboard
pages/home.hbs
News Feed
Leaderboard
View Profile
Click Asset
Get Holdings
Fetch News
News Feed
pages/feed.hbs
Leaderboard
pages/leaderboard.hbs
User Profile
pages/profile.hbs
Trade Interface
pages/asset.hbs
Backend API - Flask
POST /buy
Execute Buy
POST /sell
Execute Sell
Fetch News
Fetch Chart
Buy/Sell
GET /holdings/:user_id
User Portfolio
GET /news/:symbol
Stock News
GET /intraday/:symbol
Chart Data
GET /quote/:symbol
Stock Quotes
Query
Query
Query
Database - PostgreSQL
Transactions Table
buy/sell records
Users Table
id, username, balance
Holdings Table
user_id, symbol, qty
External APIs
TwelveData API
Historical Charts
Finnhub API
Quotes & News

Challenges

- Improving speed of stock information on feed page
- Minimizing rate/query usage for API
- Frequent problems with connecting to the database

Future Scope

- Increasing user-to-user interaction (e.g. messages)
- Allowing user to search for specific stock information and able to pin specific graphs in Feed
- Ability to take more money if you run out.
- Add option calls and shorts